

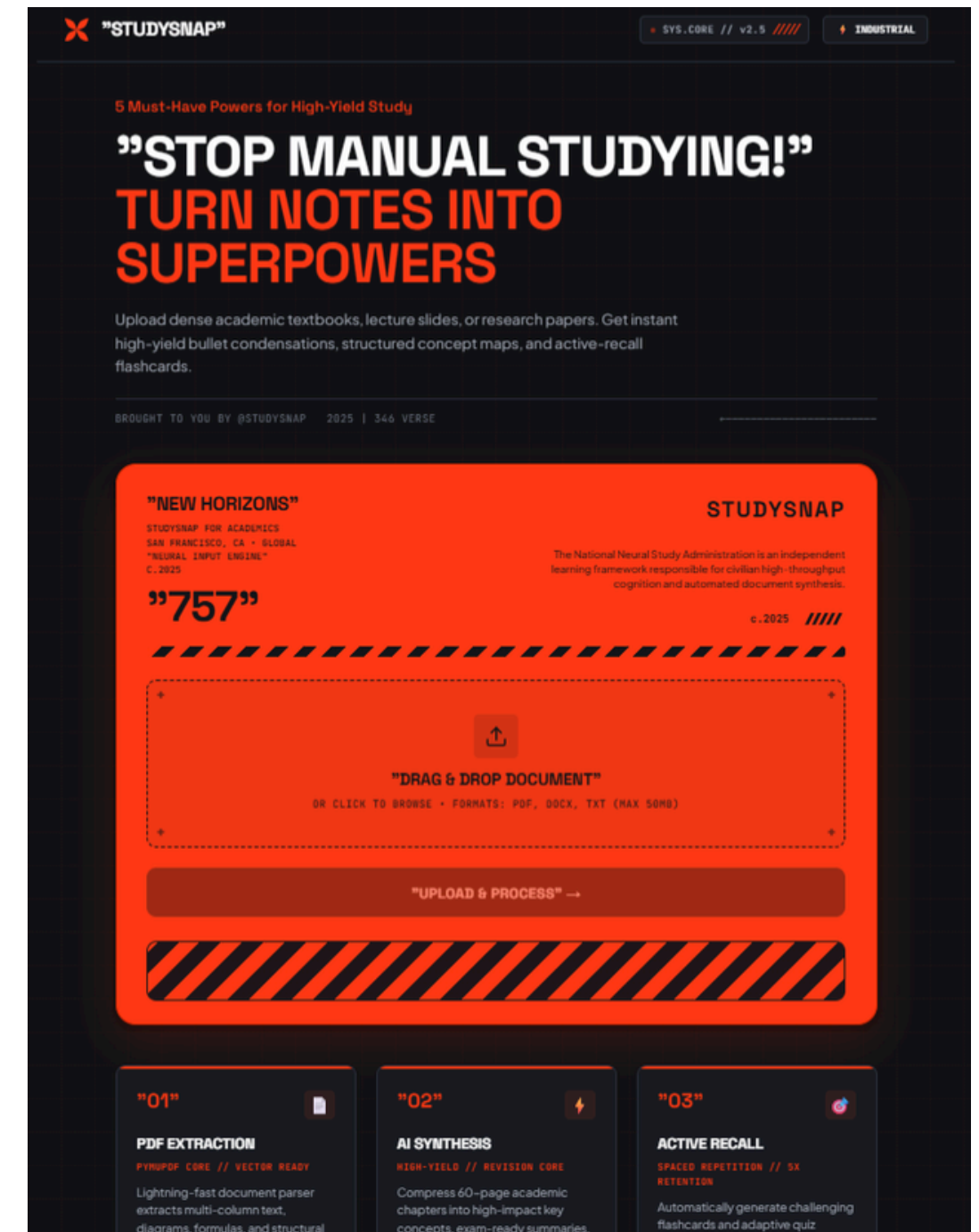
SNAPDRAGON BUILD AND PRESENT CHALLENGE

PROPOSED IDEA : **STUDYSNAP**

An AI-Powered study companion for Snapdragon-Powered PCs

Table of Contents :

1. The problem and solution
2. Working
3. Why Snapdragon
4. Current vs Future Architecture
5. How to optimise Snapdragon
6. Prototype and roadmap

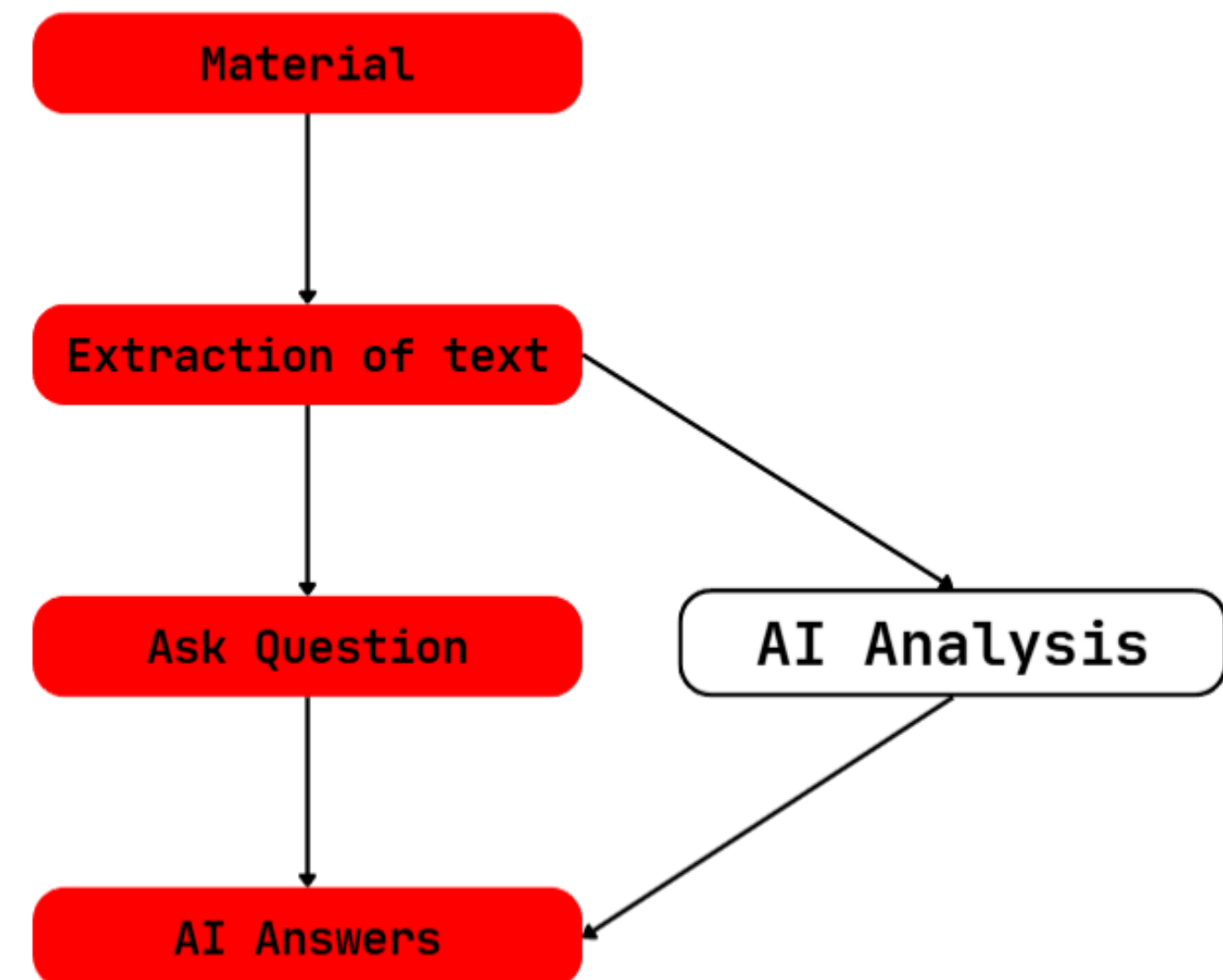


THE **PROBLEM** AND THE **SOLUTION**

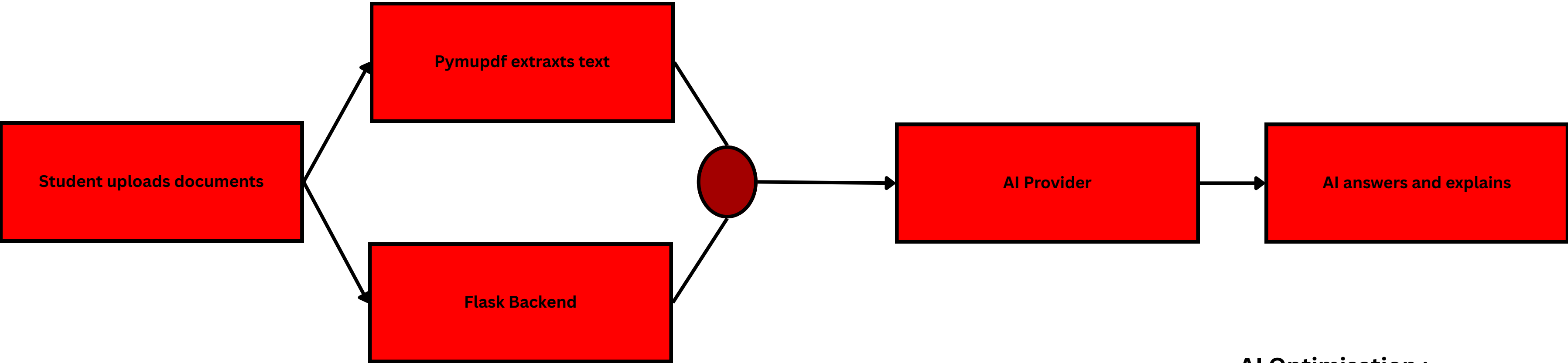
Students normally study from large pdfs and docs but getting useful information is extremely slow.

- Difficult to navigate large pdfs.
- Generic AI tools can't understand students notes.
- Cloud based AI tools have a dependency on the internet.
- Students require explanations rather than just information.

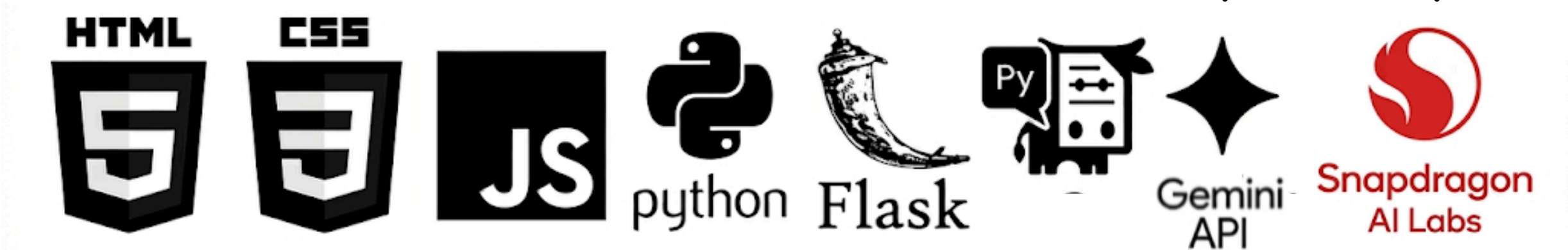
Turns students' material into an interactive one on one study session to ensure proper understanding of concepts.



HOW STUDYSNAP WORKS



TECHNOLOGY USED :



AI Optimisation :
Prototype has been designed with a replaceable AI layer so that it can evolve.

WHY **SNAPDRAGON**

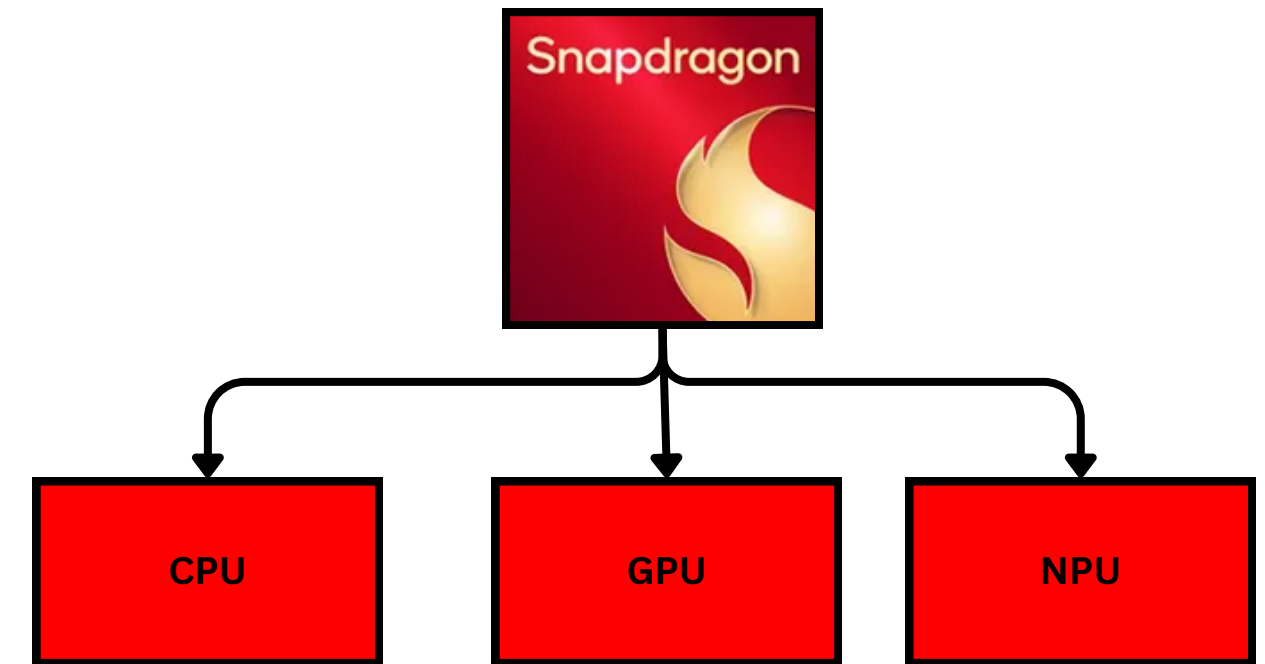
The primary goal is to move the AI assistant from the cloud to the student's PCs.

Snapdragon PCs provide :

CPU : Central Processing unit

GPU : Graphic Processing unit

NPU : Neural Processing unit

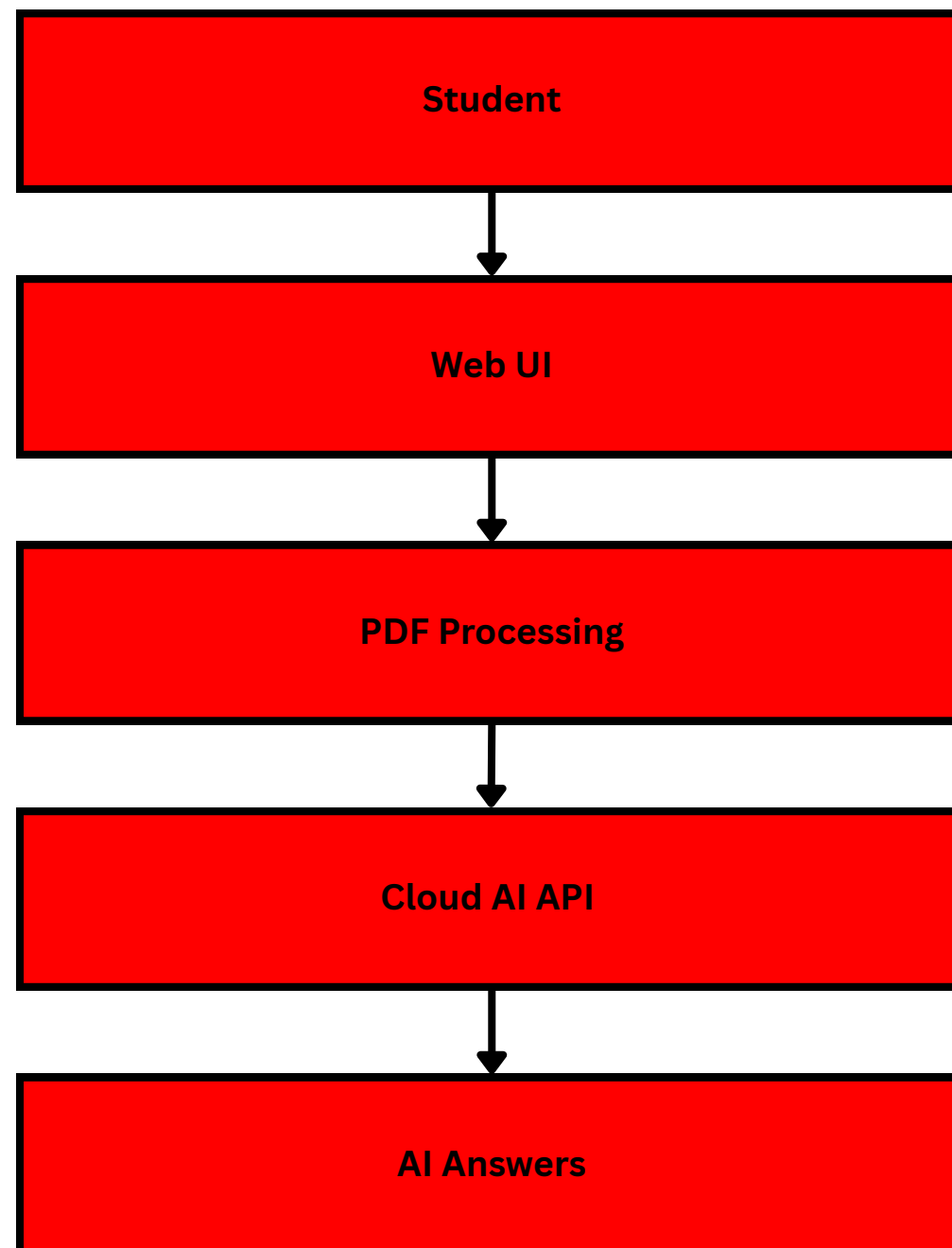


While CPU and GPU are fairly efficient for running AIs, NPU is a unit which has been specifically designed for AI/ML workloads making it the primary hardware target for StudySnap's on-device AI inference.

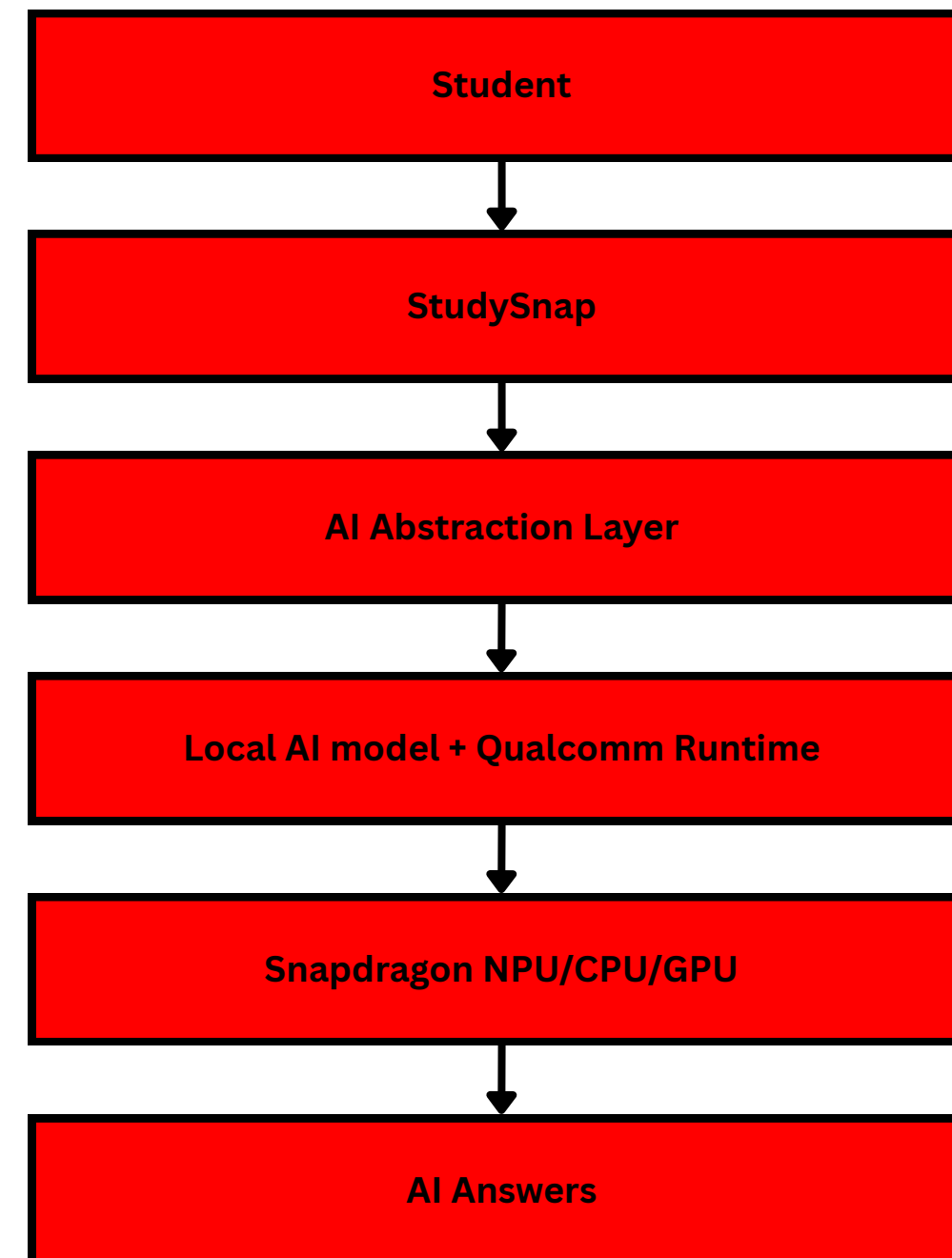
Hexagon NPU : Offers us On-device AI as well as many other features making it well suited for StudySnap.

CURRENT VS FUTURE ARCHITECTURE

Uses a Cloud API key :



Local AI model and Qualcomm Runtime :



FUTURE FEATURES, OPTIMISATION AND VISION

Vision :

“Aims to make AI-Assisted learning more **personal, practical and device-native**. Rather than asking one AI about anything, ask an **AI that understands** what you study directly from your PC.”

Future Features to be added :

- AI generated quizzes
- Analysing gaps and weak topics
- Adapts to student's learning style
- Voice interaction
- Mind-map generation
- Image/Problem understanding

Snapdragon Optimisation :

- Define StudySnap requirements.
- Shortlist suitable models
- Profile on qualcomm supported models.
- Measuring latency, accuracy and NPU footprint.
- Select best suited model.

(Qualcomm AI Hub can support model optimization and profiling workflows.)